Fiscal Year:	FY 2026	**Task Last Updated:**	FY 11/18/2025
PI Name:	Bailey, Susan M. Ph.D.		
Project Title:	Telomeres and the One Year Mission Project		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/31/2019	**End Date:**	12/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2027 per NSSC documentation and J. McFather/HRP (Ed., 2/17/26).		
Key Personnel Changes/Previous PI:	Collaborator/Study Coordinator: Kerry George, Senior Scientist KBR Laboratories received termination notice. Last day Oct 31, 2025.		
COI Name (Institution):	Jeevarajan, Antony Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC19K0434		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The ultimate goal of the studies proposed here is to establish temporal profiles of human telomere length dynamics and DNA damage responses of importance for maintenance of human health and performance during long-duration deep space missions. We hypothesize that telomere length dynamics (changes over time) represent a particularly relevant and informative biomarker of health for the astronauts, as it reflects the combined experiences and exposures encountered during spaceflight. That is, an astronaut's individual genetic susceptibilities, unique lifestyle stresses encountered (e.g., nutritional, psychological, physical), and particular environmental exposures (e.g., altered atmospheres, microgravity, space radiations) are all integrated and captured as changes in telomere length. Thus, the rate at which telomeres shorten provides a general measure of health that can be linked to aging, as well as to risk of developing age-related pathologies, ranging from reduced immune function and dementia, to cardiovascular disease and cancer. Importantly, functional telomeres are also essential for maintaining genomic integrity and stability, as they protect chromosomal termini from inappropriate degradation, and prevent these natural DNA ends from being recognized as broken DNA and triggering inappropriate DNA damage responses (DDRs). To identify trends in adaptations to human health and performance during long-duration low-Earth orbit, we propose telomere length and DDR/cytogenetic measures pioneered and validated in the NASA Twins Study/first One Year Mission, across the Integrated One-Year Mission Project onboard the International Space Station and the concurrent ground analog (prolonged isolation) component.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identifying interactive effects of genetic and nongenetic telomere length determinants and DNA damage responses will improve understanding of aging and aging trajectories (disease risk/health outcomes), and thereby guide future studies and development of mitigation strategies/countermeasures for improving health-span, not only in astronauts on long-duration missions, but for those on Earth, as well.
Task Progress:	Our ongoing work is demonstrating changes in human telomere length dynamics specifically associated with spaceflight, supporting telomeres as integrative biomarkers that encompass the extraordinary and combined life stressors and environmental exposures encountered. Thus, telomeres are being established as informative biomarkers of astronaut health, disease risk, and aging. Individual differences are also being observed, therefore determining responses in a larger, more diverse population of crewmembers, as proposed as part of the Complement of Integrated Protocols for Human Exploration Research (CIPHER), represents a critical next step. Astronauts live and work in an extreme environment fraught with extraordinary hazards and combined stressors, including space radiation exposure, microgravity and/or altered gravity, confinement and isolation (psychologically stressful), altered atmospheres and closed environment (biologically hostile), changed nutrition and microbiome – all in addition to intermittent bouts of acute stress; e.g., extravehicular activities (EVAs), and endurance/aerobic exercise to maintain bone and muscle mass. Considering the combination of unique stressors and chronic space radiation exposures associated with long-duration spaceflight, as well as the adverse health effects experienced by multiple physiological systems (e.g., dysregulated immunity, inflammation, infection), we proposed that monitoring telomere length dynamics and persistent DDRs (genomic instability) would be of particular relevance for astronauts because these informative biomarkers provide insight into individual health status during a mission, as well as potential implications and predictions for aging and disease risk later in life. The successful mission of Artemis 1 carried with it the hopes and dreams of humankind returning to the Moon and venturing beyond to Mars. As such missions and even space tourism become realities in the coming years, the number and diversity of space travelers will likewise increase. Thus, a better understanding of how long-duration spaceflight impacts human health is essential to maintaining individual astronaut performance during, and improving disease and aging trajectories following, future exploration missions. Findings from our previous NASA Twins Study and Telomeres investigations provided vital clues suggestive of potential mechanistic roles for chronic space radiation exposure underlying changes in telomere length dynamics and persistent DNA damage responses associated with long- (and short)-duration human spaceflight. Chronic exposure to the space radiation environment contributes to a number of fundamental biological features of spaceflight, including mitochondrial dysregulation and elevated oxidative stress, increased DNA damage, and dramatic shifts in telomere length dynamics. We proposed that in the context of chronic exposure to the space radiation environment, persistent DDRs to increased reactive oxygen species (ROS) production by dysfunctional mitochondria, elevated levels of oxidative damage and replication stress, and low dose, low dose rate high LET space radiations, conspire at continuously damaged telomeres, acting to either enhance telomerase activity, or perhaps more likely, particularly in blood lymphocytes with very low levels of telomerase activity, transiently activate telomerase-independent Alternative Lengthening of Telomeres (ALT) or ALT-like recombinational pathways that are at least partially responsible for the striking shifts in telomere length dynamics observed. Chronic exposure to the complex and changing space radiation environment is one of the primary hazards of long-duration spaceflight, particularly as astronauts venture deeper into space and completely outside of the protection the Earth provides. The mechanistic links between chronic exposure to space radiation and the telomeric and DNA damage responses we observed, as well as radiation dose-dependent decreases in white blood cell (WBC) counts post-spaceflight, provide support for the development of effective radiation mitigators and individualized countermeasures for upcoming deep space exploration missions. Furthermore, a major conclusion from our previous studies is that inter-individual differences in response to the combined stressors and extreme exposures associated with spaceflight predominates over general trends of individual factors, and highlights the critical need for personalized monitoring and precision space medicine strategies for current and future astronauts. While the definitive mechanisms involved in these processes remain elusive, we are testing a model based on our foundational findings: chronic exposure to the space radiation environment results in on-going genomic and telomeric DNA damage and instability, as well as transient activation of telomerase-dependent and/or independent pathways that occur in response to on-going chronic oxidative damage specifically to telomeres, which together with lymphocyte radiosensitivity, particularly those with short telomeres, and the resulting redistribution of leukocyte subsets, contribute to the telomere elongation observed during spaceflight. The goal of our current Telomeres 2 investigation as part of CIPHER is to assess a larger, more diverse cohort of astronauts, on various duration missions (ranging from several months to >one year), and thereby further elucidate and confirm underlying mechanisms of the dramatic changes in telomere length dynamics associated with spaceflight, as well as to provide additional insight into individual differences

	in response and outcomes (accelerated aging, increased disease risk), and guide future development of effective countermeasures and mitigation strategies. Our efforts are ongoing, with sample collection and data analyses working well for currently consented participants.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
Articles in Other Journals or Periodicals	Bailey SM, Mason CE. "What happens to astronauts' DNA in space." Biomedical Science Journal for Kids. https://www.sciencejournalforkids.org/articles/what-happens-to-astronauts-dna-in-space/ , Jan-2025
Articles in Peer-reviewed Journals	Chino Y, Olson JD, Aichele S, Bailey SM, Schaaf GW, Cline JM, Johnson TE. "The impact of dose, sex, and age at time of acute whole-body radiation exposure on long-term hematopoiesis in rhesus macaques (Macaca mulatta)." Radiat Res. 2025 Oct 1;204(4):408-420 https://doi.org/10.1667/RADE-24-00241.1 ; PubMed PMID: 40832933 , Oct-2025